

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-702(A)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Microwave Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Enumerate the basic advantages of microwaves. Briefly explain the applications of microwaves.
b) A rectangular waveguide with dimension of 3×2 cms operates in TM_{11} mode at 10 GHz. Determine the characteristic wave impedance.
2. a) What is a magic Tee junction? Derive the S matrix of a magic Tee.
b) Write short notes on:
 - i) Waveguide Irises.
 - ii) Rat Race hybrid
 - iii) Dielectric phase shifters
3. a) With a neat sketch, explain the structure and principle of operation of TWT Amplifier.
b) Draw the structure and explain the velocity modulation process in two cavity klystron amplifier.
4. a) What are the bulk properties of GUNN diode that give rise to negative resistance?

EC-702(A)-CBGS

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[2]

- b) Explain the physical structure and construction of IMPATT diodes.
5. a) Give the measurement procedure for Q factor of a resonant cavity and attenuation constant at microwave frequencies.
b) Explain the double minimum method of measuring VSWR.
6. a) Why are S - parameters used at microwave frequencies explain? Give the properties of S parameters. Derive S - matrix for series Tee using the properties of S parameters.
b) An X-band pulsed conventional magnetron has the following parameters. Anode voltage $V_o = 5.5$ kV, Beam current $I_o = 4.5$ A, Operating frequency $f = 9 \times 10^9$ Hz, Resonant conductance $G_r = 2 \times 10^{-4}$ mho, Loaded conductance $G_l = 2.5 \times 10^{-5}$ mho, Vane capacitance $C = 2.5$ PF. Compute:
i) Angular resonant frequency
ii) Unloaded quality factor
iii) Loaded quality factor
7. a) Explain the method of microwave power measurement using Bolometer.
b) Two identical 30dB directional couplers are used to sample incident and reflected power in a wave guide. VSWR = 2 and the output of the coupler sampling incident power = 4.5 mW. What is the value of reflected power?
8. a) Explain the impedance measurement using Reflectometer.
b) Why pi-mode operation is preferred in cylindrical type magnetron ? Give its working principle with neat sketches.
